

KHOMUTOVO, Russian Federation

WMO: 321500

Lat: **46.889N**Lon: **142.718E**Elev: **59**StdP: **14.66**Time Zone: **11.00 (E11)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-10.9	-19.9	1.9	-15.1	-14.7	2.5	-10.6	39.1	20.6	30.1	20.7	2.1	350	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.8	78.8	68.4	75.5	66.8	73.0	65.4	70.8	75.8	69.0	73.2	67.0	71.0	8.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	108.4	73.7	67.6	101.6	71.6	65.7	94.9	69.7	34.8	76.1	33.2	73.0	31.6	71.0	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	19.7	17.4	-23.3	85.1	4.2	3.1	-26.4	87.4	-28.8	89.2	-31.2	91.0	-34.3	93.3		
(5)			-23.5	74.5	4.2	2.3	-26.6	76.1	-29.0	77.5	-31.4	78.8	-34.5	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	9.5	10.4	21.6	35.2	46.2	54.2	61.5	63.8	56.4	43.8	29.6	15.4
	DBStd	20.31	9.31	9.07	8.13	6.13	6.38	5.79	5.26	5.22	5.92	6.18	8.51	8.48
	HDD50	5777	1256	1108	881	444	154	25	0	0	12	210	614	1073
	HDD65	10143	1721	1528	1346	894	584	327	133	88	263	658	1063	1538
	CDD50	1191	0	0	0	0	36	150	357	427	204	16	1	0
	CDD65	82	0	0	0	0	0	2	25	50	5	0	0	0
	CDH74	451	0	0	0	0	9	30	124	251	37	0	0	0
	CDH80	53	0	0	0	0	2	2	11	36	2	0	0	0
Wind	WSAvg	7.0	7.4	6.8	7.6	7.8	8.3	7.5	6.5	5.9	5.9	6.6	6.5	6.6
Precipitation	PrecAvg	34.6	2.1	1.6	1.7	2.3	2.6	2.3	3.5	4.1	4.1	3.9	3.0	2.6
	PrecMax	49.1	5.4	4.2	4.3	5.5	5.3	6.9	9.0	12.6	8.8	8.5	5.6	5.9
	PrecMin	25.8	0.2	0.0	0.3	0.5	0.4	0.0	0.6	0.4	0.4	1.2	0.3	0.5
	PrecStd	5.8	1.1	1.0	0.9	1.1	1.3	1.4	2.0	2.4	2.0	1.8	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	37.7	43.1	61.0	72.9	77.9	81.2	84.2	78.6	66.6	55.5	40.6
		MCWB	31.9	33.2	36.1	47.8	56.6	64.4	69.6	70.7	66.4	56.4	48.4	38.9
	2%	DB	30.5	33.5	39.2	55.5	67.6	73.3	77.3	80.1	73.7	62.3	50.3	35.8
		MCWB	29.4	31.1	34.0	45.3	54.5	62.3	68.2	69.9	63.9	53.9	45.7	34.2
	5%	DB	28.3	29.9	36.0	50.2	63.0	69.6	73.9	76.8	70.6	58.9	46.5	32.7
		MCWB	26.9	27.9	32.1	42.5	51.9	60.3	66.4	68.6	62.5	52.4	43.7	31.3
	10%	DB	25.2	26.6	33.9	46.0	59.0	66.0	71.3	73.3	67.8	55.7	42.9	30.3
		MCWB	23.7	24.5	30.6	39.7	49.9	58.4	65.3	67.2	61.2	50.5	39.6	28.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.2	34.4	37.7	50.3	58.5	66.9	72.2	74.2	69.0	59.1	50.3	38.7
		MCDB	32.7	36.2	40.7	60.0	69.3	74.5	78.3	80.7	74.3	63.2	53.0	39.5
	2%	WB	29.7	31.0	34.9	45.9	55.8	63.5	70.0	71.9	66.5	56.3	46.9	34.4
		MCDB	30.5	32.8	37.8	53.2	64.9	71.0	75.3	76.9	71.0	59.8	49.2	35.2
	5%	WB	26.8	27.6	32.9	42.9	53.2	61.1	67.8	70.0	64.4	53.9	43.4	31.6
		MCDB	27.7	29.4	35.4	49.3	61.3	67.8	72.2	74.0	68.9	57.6	45.8	32.5
	10%	WB	23.8	24.4	31.1	40.2	50.8	59.0	65.9	68.4	62.3	51.6	39.8	28.5
		MCDB	25.3	26.3	33.3	45.0	57.5	64.5	69.8	72.1	66.4	55.0	42.4	30.0
Mean Daily Temperature Range	5% DB	MDBR	21.2	24.4	19.7	16.9	19.4	17.6	15.3	15.8	20.3	20.1	16.5	19.1
		MCDBR	16.8	24.2	20.1	25.4	28.3	26.2	21.1	21.7	24.0	24.9	21.7	17.8
		MCWBR	15.9	21.3	15.7	16.0	16.1	15.7	12.7	12.8	15.5	17.6	17.7	16.7
	5% WB	MCDBR	14.9	22.3	17.7	23.6	25.7	22.7	17.9	17.5	19.9	21.2	19.9	16.6
		MCWBR	14.6	20.0	14.8	15.5	15.5	14.4	11.7	11.3	13.5	16.4	17.2	16.2
Clear-Sky Solar Irradiance	taub		0.313	0.355	0.415	0.492	0.520	0.494	0.462	0.440	0.397	0.403	0.374	0.336
	taud		2.045	1.898	1.823	1.810	1.872	2.046	2.213	2.287	2.340	2.152	2.109	2.045
	Ebn at Noon		232	244	247	241	240	248	255	256	257	229	207	204
	Edh at Noon		31	45	56	63	61	52	43	39	34	35	30	28
All-Sky Solar Radiation	RadAvg		401	668	1027	1434	1538	1513	1404	1300	1170	787	440	313
	RadStd		45	67	98	150	138	173	173	115	122	49	38	26